Classical and Bayesian Estimation of the Lomax Distribution under Joint Improved Adaptive Progressive Censored Competing Risks Data

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Abstract

In this article, we consider the classical and Bayesian estimation of the Lomax distribution when two production lines are tested together under a joint improved adaptive progressive Type-II censoring scheme (JIAPCS) and every failure is recorded with its cause (two independent causes per line). We write down the likelihood, show that it splits by population, and derive the maximum likelihood estimators (MLEs): the two shape parameters have closed forms for a given scale parameter, a finite MLE of the scale parameter exists under a simple sample condition (otherwise the likelihood is maximised in the exponential limit), and an order-restricted MLE for $\alpha_{d1} \leq \alpha_{d2}$ is obtained in closed form. Bayes estimates under squared-error, LINEX and general-entropy losses, and HPD credible intervals, are computed without and with the order restriction using a collapsed Metropolis–Gibbs sampler. The performance of different estimators is evaluated through extensive numerical simulations (1000 replications per scenario) in terms of mean-squared error and absolute bias, while ACI and HPD credible intervals are assessed in terms of average width and coverage probability. We find that the Lomax MLE exists in only 51–79% of the samples and is unstable when it does, that proper-prior Bayes estimates are far more accurate, and that the order restriction helps little when true but hurts when violated. Finally, two real data sets (aircraft air-conditioning failures and electrical appliances) are analyzed for illustration purposes; both are close to exponential, which shows the limits of the Lomax model.

Keywords: Lomax distribution; competing risks; joint progressive censoring; gamma prior; maximum likelihood estimator; Bayesian estimation.

1 Introduction

Reliability testing is one of the important methods to assess the failure pattern of different products. In such situations, items are put on test under some operational condition and then failure times are recorded. A unit can often fail for more than one reason (for example a compressor fault or any other fault in an air-conditioning unit). The causes compete: we see only the first one that acts, together with its identity. This is the competing risks model, which goes back to Cox [1]; for the Lomax lifetime under progressive censoring see Cramer and Schmiedt [2].

Life tests are costly, so they are stopped early. In progressive Type-II censoring a fixed number of surviving units is withdrawn at each failure. To compare the products of two production lines, one can use both samples in a single test. Balakrishnan and Rasouli [3] proposed joint progressive censoring, and Mondal and Kundu [4] introduced the balanced version (BJPCS), in which both samples lose the same number of units after every failure. Chen and Gui [5] studied BJPCS with competing risks for the Burr-XII distribution, and this is the paper our project follows. A weakness of progressive schemes is that the test can last very long. Ng, Kundu and Chan [6] let the scheme adapt after a time threshold T_1, and Yan, Li and Yu [7] added a second, hard deadline T_2; this is the improved adaptive scheme. Competing risks under improved adaptive censoring have been

studied for several lifetime models (see e.g. [8]), but we did not find a treatment for the Lomax distribution, nor for the two-population (joint) version of the scheme.
The Lomax (Pareto II) distribution [9] has the decreasing hazard $h(x) = \alpha\lambda/(1+\lambda x)$ and a heavy tail, which suits products with early failures. It contains the exponential distribution as the limit $\alpha \to \infty$, $\lambda \to 0$ with $\alpha\lambda$ fixed, and this limit will turn out to matter a lot.
Contributions. (i) We formulate the likelihood of the Lomax competing risks model under JIAPCS. (ii) We obtain the MLEs, with and without the order restriction $\alpha_{d1} \le \alpha_{d2}$, and study when they exist. (iii) We derive Bayes estimates under three loss functions and HPD intervals, with independent gamma priors and with the truncated (order-restricted) prior. (iv) We compare everything by simulation and (v) apply it to two real data sets. Along the way we try to derive the main results from scratch using only tools from the course (MLE, Fisher information, Cramér–Rao, likelihood ratio tests, Bayes estimators under a loss function; see e.g. Casella and Berger [10]).
The organization of the paper is as follows. The model, the censoring scheme and the likelihood are set up in Section 2, which also gives the maximum likelihood estimators, their existence, the order-restricted MLE and the interval estimates. Bayesian estimation, with and without the order restriction, is developed in Section 3. Section 4 presents a simulation study and two real data applications. Finally, the conclusion is given in Section 5.

2 Maximum Likelihood Estimation

Model. Population Ω_d $(d = 1, 2)$ has m_d units. Each unit has two latent lifetimes W_{d1}, W_{d2}, independent, with

$$S_{dr}(x) = (1+\lambda_d x)^{-\alpha_{dr}}, \qquad h_{dr}(x) = \frac{\alpha_{dr}\lambda_d}{1+\lambda_d x}, \qquad x > 0,\ r = 1, 2.$$

The two causes in a population share the scale-type parameter λ_d but may have different shapes. This common-λ_d assumption is what makes the algebra below work. A unit fails at $\min(W_{d1}, W_{d2})$, whose survival function is $S_d(x) = (1+\lambda_d x)^{-(\alpha_{d1}+\alpha_{d2})}$ (multiply the two survival functions). Because $h_{d1}/(h_{d1}+h_{d2}) = \alpha_{d1}/(\alpha_{d1}+\alpha_{d2})$ does not depend on x, the cause is r with probability $\alpha_{dr}/(\alpha_{d1}+\alpha_{d2})$, independently of the failure time. The six unknowns are $\theta = (\alpha_{11}, \alpha_{12}, \lambda_1, \alpha_{21}, \alpha_{22}, \lambda_2)$.

The JIAPCS. Fix $k < \min(m_1, m_2)$, removal numbers $R_1, \dots, R_{k-1} \ge 0$ with $\sum_{i<k}(R_i+1) < \min(m_1, m_2)$, and thresholds $T_1 < T_2$. Both samples are put on test together. At the i-th failure, at time $w_{(i)}$, in population Ω_z: (1) (adaptive rule) the number of withdrawals is R_i if $w_{(i)} < T_1$ and 0 if $w_{(i)} \ge T_1$; (2) (balanced rule) R_i survivors are withdrawn at random from Ω_z and $R_i + 1$ from the other population, so both populations lose R_i+1 units; (3) (improved rule) the test ends at the k-th failure, or at T_2 if fewer than k failures have happened by then; all survivors are withdrawn at the end. Three cases occur: I ($w_{(k)} < T_1$, the ordinary balanced scheme), II ($T_1 \le w_{(k)} < T_2$, removals stop after T_1) and III ($w_{(k)} \ge T_2$, the test is stopped at T_2 with fewer than k failures). The data are the failure times together with the population and cause of each failure.

Likelihood. Let $t_{(1)} < \cdots < t_{(K)}$ be the event times of the test (failure times and the final time) and let ν_{dj} be the number of Ω_d units that fail or are withdrawn at $t_{(j)}$, so $\sum_j \nu_{dj} = m_d$. Let $x_{d1}, \dots, x_{dN_d}$ be the failure times in Ω_d with causes $\delta_{di} \in \{1, 2\}$, and $N_{dr} = \#\{i : \delta_{di} = r\}$, $N_d = N_{d1} + N_{d2}$. A unit that fails at x from cause r contributes $f_{dr}(x)S_{dr'}(x) = h_{dr}(x)S_d(x)$ (cause r acts at x and the other cause has not yet acted). A withdrawn unit at t contributes $S_d(t)$. The decisions of the scheme depend only on the past, so they are non-informative and the likelihood is proportional to the product of these terms. Taking logs and collecting the survival factors of failed

and withdrawn units in ν_{dj},

$$\ell_d = N_{d1}\ln\alpha_{d1} + N_{d2}\ln\alpha_{d2} + N_d\ln\lambda_d - \sum_{i=1}^{N_d}\ln(1+\lambda_d x_{di}) - (\alpha_{d1}+\alpha_{d2})D_d(\lambda_d), \tag{1}$$

where $D_d(\lambda) = \sum_{j=1}^{K}\nu_{dj}\ln(1+\lambda t_{(j)})$. The full log-likelihood is $\ell_1+\ell_2$ and the two parts share no parameters, so Ω_1 and Ω_2 are analysed separately with the same three-parameter routine. From now on we drop the index d.

2.1 MLE Under no order restriction

Theorem 2.1 (Shape parameters)**.** *For fixed λ and $N_1, N_2 \geq 1$, ℓ is maximised by $\hat{\alpha}_r(\lambda) = N_r/D(\lambda)$, $r = 1, 2$.*

Proof. As a function of the shapes, $\ell = \sum_r N_r\ln\alpha_r - (\alpha_1+\alpha_2)D + \text{const}$. Setting $\partial\ell/\partial\alpha_r = N_r/\alpha_r - D = 0$ gives $\hat{\alpha}_r = N_r/D$, and $\partial^2\ell/\partial\alpha_r^2 = -N_r/\alpha_r^2 < 0$ shows it is a maximum. □

Substituting back gives the profile log-likelihood of λ (up to a constant),

$$\ell_p(\lambda) = N\ln\lambda - \sum_{i=1}^{N}\ln(1+\lambda x_i) - N\ln D(\lambda), \tag{2}$$

with score $\ell_p'(\lambda) = N/\lambda - \sum_i x_i/(1+\lambda x_i) - ND'(\lambda)/D(\lambda)$ and $D'(\lambda) = \sum_j \nu_j t_{(j)}/(1+\lambda t_{(j)})$.

Theorem 2.2 (Existence of $\hat{\lambda}$)**.** *Let $S_1 = \sum_i x_i$, $W = \sum_j \nu_j t_{(j)}$, $Q = \sum_j \nu_j t_{(j)}^2$ and $\Psi = NQ/(2W) - S_1$. If $\Psi > 0$, then ℓ_p has a maximum at a finite $\hat{\lambda} \in (0,\infty)$ which solves $\ell_p'(\lambda) = 0$, and $\hat{\alpha}_r = N_r/D(\hat{\lambda})$ are the MLEs. If ℓ_p is maximised only at the boundary, the supremum of the likelihood is approached as $\lambda \to 0$ with $\alpha_r \to \infty$ and $\alpha_r\lambda \to N_r/W$, i.e. by the exponential competing risks model.*

Proof. (a) Behaviour near 0. Use $\ln(1+u) = u - u^2/2 + O(u^3)$. Then $D(\lambda) = \lambda W - \frac{1}{2}\lambda^2 Q + O(\lambda^3)$, so $\ln D = \ln\lambda + \ln W + \ln\big(1 - \frac{\lambda Q}{2W} + O(\lambda^2)\big) = \ln\lambda + \ln W - \frac{\lambda Q}{2W} + O(\lambda^2)$, and $\sum_i \ln(1+\lambda x_i) = \lambda S_1 + O(\lambda^2)$. Putting these into (2), the $N\ln\lambda$ terms cancel and

$$\ell_p(\lambda) = -N\ln W + \lambda\Big(\frac{NQ}{2W} - S_1\Big) + O(\lambda^2) = -N\ln W + \lambda\Psi + O(\lambda^2).$$

So $\ell_p'(0+) = \Psi$, and if $\Psi > 0$ the profile is increasing near 0. *(b) Behaviour at* ∞. $N\ln\lambda - \sum\ln(1+\lambda x_i) = -\sum\ln(x_i + 1/\lambda) \to -\sum\ln x_i$ is finite, while $D(\lambda) \geq \ln(1+\lambda t_{(1)}) \to \infty$ (the weights ν_j add up to $m > 0$). Hence $-N\ln D \to -\infty$ and $\ell_p \to -\infty$. *(c)* ℓ_p is continuous on $(0,\infty)$, starts above its limit $-N\ln W$ (by (a)) and goes to $-\infty$ (by (b)), so it has a global maximum at an interior point where $\ell_p' = 0$. For the boundary statement let $\lambda \to 0$ with $\alpha_r\lambda = \kappa_r$ fixed. Then $(1+\lambda x)^{-\alpha_r} \to e^{-\kappa_r x}$, $\sum\ln(1+\lambda x_i) \to 0$, $(\alpha_1+\alpha_2)D \to (\kappa_1+\kappa_2)W$, so (1) tends to $\sum_r N_r\ln\kappa_r - (\kappa_1+\kappa_2)W$, which is the exponential competing risks log-likelihood and is maximised at $\kappa_r = N_r/W$. □

Remark 2.1 (About uniqueness and non-existence)**.** *$\Psi > 0$ is only a sufficient condition in the proof. For a complete sample ($\nu_j = 1$, $W = S_1$) it reads $\overline{x^2} > 2\bar{x}^2$, i.e. the squared coefficient of variation is larger than 1; so the Lomax likelihood needs a clearly heavy-tailed sample. We could not prove uniqueness of the root, so we checked it numerically: in 1000 simulated samples (design D2, "mid" pattern, thresholds PB, Ω_1) none had two or more interior maxima of ℓ_p, 65.7% had exactly one and 34.3% had none, and the event $\{\Psi > 0\}$ coincided with the event that a finite MLE exists. So in practice $\Psi > 0$ is necessary and sufficient and the root is unique. A claim of a unique finite MLE for every sample would be wrong: when $\Psi \leq 0$ the MLE does not exist (flat profile, Fig. 2(a)).*

Computation. We evaluate ℓ_p on a log-grid of 400 values of λ around N/W, take the grid maximum and refine the root of ℓ_p' by bisection (60 steps). If the grid maximum is on the boundary we declare that the MLE does not exist.

Remark 2.2 (A check in the simplest case)**.** *Take one cause, a complete sample of size n and known λ. Then $Y = \ln(1+\lambda X)$ satisfies $P(Y > y) = P\big(X > (e^y-1)/\lambda\big) = e^{-\alpha y}$, so $Y \sim \mathrm{Exp}(\alpha)$ and $T = \sum_i Y_i \sim \mathrm{Gamma}(n,\alpha)$ (rate α); T is sufficient and complete, and the MLE is $\hat\alpha = n/T$. Since $E(1/T) = \alpha/(n-1)$, we get $E\hat\alpha = n\alpha/(n-1)$: the MLE is biased in small samples. The estimator $\tilde\alpha = (n-1)/T$ is unbiased and a function of a complete sufficient statistic, hence the UMVUE (Lehmann–Scheffé) with $\mathrm{Var}(\tilde\alpha) = \alpha^2/(n-2)$, which is slightly above the Cramér–Rao bound α^2/n (the bound is not attained). A simulation with $n = 10$, $\alpha = 0.8$, $\lambda = 1.5$ (2×10^5 replications) gave mean 0.889 for $\hat\alpha$ (theory 0.889) and mean 0.800, variance 0.0796 for $\tilde\alpha$ (theory 0.0800). Moreover $\alpha T \sim \mathrm{Gamma}(n,1)$ is a pivot, which gives the exact interval $\big[\chi^2_{2n,0.025}/(2T),\ \chi^2_{2n,0.975}/(2T)\big]$, and by monotone likelihood ratio the UMP test of $H_0 : \alpha \le \alpha_0$ against $H_1 : \alpha > \alpha_0$ rejects for small T. Under JIAPCS the role of T is played by $D(\lambda)$, which explains the form N_r/D in Theorem 2.1, but D is no longer exactly Gamma (the event times are order statistics of the pooled test), so we use asymptotic and bootstrap intervals below.*

2.2 MLE Under order restriction

Often one cause is known (or believed) to be the more dangerous one, so we impose $\alpha_{d1} \le \alpha_{d2}$.

Theorem 2.3 (Restricted MLE)**.** *Under $\alpha_1 \le \alpha_2$ the restricted MLEs are*

$$\hat\alpha_1^R = \frac{\min(N_1, N/2)}{D(\hat\lambda)}, \qquad \hat\alpha_2^R = \frac{\max(N_2, N/2)}{D(\hat\lambda)},$$

with the same $\hat\lambda$ as in the unrestricted case. So the restricted estimate equals the unrestricted one if $N_1 \le N_2$, and pools the two shapes to $N/(2D(\hat\lambda))$ otherwise.

Proof. Fix λ and write $g(\alpha_1,\alpha_2) = N_1 \ln\alpha_1 + N_2 \ln\alpha_2 - (\alpha_1+\alpha_2)D$, a concave function with unrestricted maximiser $(N_1,N_2)/D$. If $N_1 \le N_2$ this point satisfies $\alpha_1 \le \alpha_2$, so it is also the restricted maximiser. If $N_1 > N_2$ it lies outside the convex region $\{\alpha_1 \le \alpha_2\}$, and the maximum of a concave function over a convex region is then on the boundary $\alpha_1 = \alpha_2 = \alpha$, where $g = N\ln\alpha - 2\alpha D$ is maximised at $\alpha = N/(2D)$. In both cases the maximised value is $-N\ln D$ plus a quantity free of λ (it equals $\sum N_r \ln N_r - N$ or $N\ln(N/2) - N$), so the profile in λ is (2) up to a constant and has the same maximiser $\hat\lambda$. The compact form follows from $\min(N_1, N/2)$ and $\max(N_2, N/2)$. □

Is the restriction plausible? A likelihood ratio test. To test $H_0 : \alpha_1 = \alpha_2$ against $\alpha_1 \ne \alpha_2$, note from the proof above that both fits have the same $\hat\lambda$, so the maximised log-likelihoods differ only through the α-part:

$$\Lambda = 2\Big[N_1 \ln N_1 + N_2 \ln N_2 - N\ln(N/2)\Big] = 2\Big[N_1 \ln\frac{2N_1}{N} + N_2 \ln\frac{2N_2}{N}\Big],$$

which does not depend on λ and is asymptotically χ^2_1 under H_0 (Wilks). It is just the likelihood ratio test of "$p = 1/2$" for the binomial cause counts, as it should be, since the cause is independent of the time. We use it in Sec. 4.2.

2.3 Interval estimation

For $\vartheta = (\alpha_1, \alpha_2, \lambda)$ the observed information is minus the Hessian of (1). Differentiating twice, using $D''(\lambda) = -E(\lambda)$ with $E = \sum_j \nu_j t_{(j)}^2/(1+\lambda t_{(j)})^2$,

$$I(\hat\vartheta) = \begin{bmatrix} N_1/\hat\alpha_1^2 & 0 & D' \\ 0 & N_2/\hat\alpha_2^2 & D' \\ D' & D' & \dfrac{N}{\hat\lambda^2} - \sum_i \dfrac{x_i^2}{(1+\hat\lambda x_i)^2} - (\hat\alpha_1+\hat\alpha_2)E \end{bmatrix}.$$

We checked these formulas against numerical second differences (maximum difference 4.5×10^{-6}). By asymptotic normality (the inverse information approximates the Cramér–Rao bound for large samples), $\hat\vartheta_l \pm z_{0.975}\sqrt{[I^{-1}]_{ll}}$ is the ACI. To keep the lower limit positive we also use the log-transformed ACI: by the delta method $\mathrm{Var}(\ln\hat\vartheta_l) \approx \mathrm{Var}(\hat\vartheta_l)/\hat\vartheta_l^2$, giving $\big(\hat\vartheta_l\, e^{-z\,se/\hat\vartheta_l},\ \hat\vartheta_l\, e^{z\,se/\hat\vartheta_l}\big)$. The parametric bootstrap-p interval generates B JIAPCS samples from the fitted model (Algorithm 2), recomputes the MLE in those where it exists, and takes the 2.5% and 97.5% percentiles.

3 Bayesian Estimation

Estimators as decision problems. Let $\pi(\vartheta \mid \text{data})$ be the posterior and $L(\vartheta, a)$ a loss function. The Bayes estimator minimises the posterior expected loss $E[L(\vartheta,a) \mid \text{data}]$, which also minimises the Bayes risk. Three cases are used.

- *Squared error (SE):* $L = (a-\vartheta)^2$; setting $\frac{d}{da}E(a-\vartheta)^2 = 2(a - E\vartheta) = 0$ gives the posterior mean $\hat\vartheta_{SE} = E(\vartheta \mid \text{data})$.
- *LINEX* [11], $p \neq 0$: $L = e^{p(a-\vartheta)} - p(a-\vartheta) - 1$; $\frac{d}{da}EL = p\,e^{pa}E(e^{-p\vartheta}) - p = 0$ gives $\hat\vartheta_L = -\frac{1}{p}\ln E(e^{-p\vartheta} \mid \text{data})$.
- *General entropy (GE)* [12], $q \neq 0$: $L = (a/\vartheta)^q - q\ln(a/\vartheta) - 1$; $\frac{d}{da}EL = q\,a^{q-1}E(\vartheta^{-q}) - q/a = 0$ gives $\hat\vartheta_{GE} = \big[E(\vartheta^{-q} \mid \text{data})\big]^{-1/q}$.

Positive p, q penalise over-estimation more and give smaller estimates. We use $p, q \in \{-0.5, 0.5\}$. Given posterior draws $\vartheta^{(1)}, \ldots, \vartheta^{(M)}$ the expectations are replaced by averages. The HPD interval is the shortest interval containing 95% of the draws [13].

3.1 Bayesian Estimation under No Order Restriction

3.1.1 Prior Assumptions

For each population we take independent gamma priors (shape–rate form) $\alpha_{dr} \sim \mathrm{Gamma}(a_{dr}, b_{dr})$ and $\lambda_d \sim \mathrm{Gamma}(c_d, e_d)$. Three prior sets are used: IP (informative: mean equal to the true value, variance mean/4, i.e. shape = 4×true and rate 4; used only in simulation, where the truth is known), WIP (weakly informative, all hyperparameters equal to 1) and NIP (all hyperparameters 10^{-3}).

Posterior and sampler. Write $s_r = a_r + N_r$ and $r_r(\lambda) = b_r + D(\lambda)$. Multiplying the likelihood $\exp(\ell)$ by the priors,

$$\pi(\alpha_1, \alpha_2, \lambda \mid \text{data}) \propto \prod_{r=1}^{2} \alpha_r^{s_r-1} e^{-\alpha_r r_r(\lambda)} \cdot \lambda^{c+N-1} e^{-e\lambda} \prod_i (1+\lambda x_i)^{-1}.$$

Given λ, the shapes are independent $\mathrm{Gamma}(s_r, r_r(\lambda))$. Integrating them out with $\int_0^\infty \alpha^{s-1}e^{-\alpha r}d\alpha = \Gamma(s)r^{-s}$ gives the marginal posterior of λ,

$$\pi(\lambda \mid \text{data}) \propto \lambda^{c+N-1} e^{-e\lambda} \prod_i (1+\lambda x_i)^{-1} \prod_{r=1}^{2} \big(b_r + D(\lambda)\big)^{-s_r}. \tag{3}$$

Because the shapes are integrated out, only a one-dimensional chain is needed (Algorithm 1). We validated the sampler against numerical integration of (3): posterior means of λ agreed within Monte Carlo error (e.g. 1.237 vs 1.221 unrestricted and 1.052 vs 1.028 restricted).

Algorithm 1 Collapsed Metropolis–Gibbs sampler

1: Start at $\lambda^{(0)} = \hat{\lambda}$ (or N/W).
2: **for** $t = 1, \ldots, T$ **do**
3: Propose $\lambda^* = \lambda^{(t-1)} e^{\sigma Z}$, $Z \sim N(0,1)$; accept with probability $\min\{1, \pi(\lambda^*)\lambda^*/[\pi(\lambda^{(t-1)})\lambda^{(t-1)}]\}$ (the factor λ comes from the log-scale proposal; σ is tuned in burn-in to about 35% acceptance).
4: Draw $\alpha_r^{(t)} \sim \text{Gamma}(s_r, r_r(\lambda^{(t)}))$ independently. (restricted case: see Section 3.2)
5: **end for**

Remark 3.1 (Vague priors fail for the Lomax model)**.** *When the likelihood is nearly flat along the ridge $\alpha\lambda \approx$ constant (the exponential limit), a vague prior puts a lot of posterior mass on huge α. With the NIP prior on 500 samples from design D1 (Ω_1), the average posterior mean of $(\alpha_{11}, \alpha_{12}, \lambda_1)$ was $(39.8, 79.1, 3.07)$ against the true $(0.5, 1.0, 1.0)$, the MSE of the posterior mean of α_{11} was 1968, and the LINEX estimate with negative p was infinite in 98.4% of samples because $E[e^{0.5\alpha} \mid \text{data}]$ diverges. Posterior medians were better $(3.9, 8.2, 0.92)$ but still poor. We therefore use proper, moderately informative priors (WIP and IP).*

3.2 Bayesian Estimation under Order Restriction

3.2.1 Prior Assumptions

For the restricted case the joint prior of the shapes is the product of the gamma densities truncated to the region $\alpha_{d1} < \alpha_{d2}$,

$$\pi_R(\alpha_{d1}, \alpha_{d2}) \propto \pi(\alpha_{d1})\pi(\alpha_{d2}) \mathbb{1}\{\alpha_{d1} < \alpha_{d2}\},$$

and λ_d keeps its gamma prior (the normalising constant of the truncated prior does not involve λ or the data). Given λ, the restricted conditional posterior of (α_1, α_2) is the product of the two $\text{Gamma}(s_r, r_r)$ densities restricted to $\alpha_1 < \alpha_2$. Integrating over that region instead of over the whole quadrant multiplies (3) by

$$P_\lambda = \Pr(G_1 < G_2), \qquad G_r \sim \text{Gamma}(s_r, r_r) \text{ independent.}$$

This probability is explicit: with $X_r = r_r G_r \sim \text{Gamma}(s_r, 1)$ we have $G_1 < G_2 \iff X_1/(X_1 + X_2) < \rho$ with $\rho = r_1/(r_1 + r_2)$ and $X_1/(X_1 + X_2) \sim \text{Beta}(s_1, s_2)$, so $P_\lambda = I_\rho(s_1, s_2)$, the regularised incomplete beta function. The same chain as in Algorithm 1 is run with this extra factor, and step 4 is replaced by an exact draw: $B \sim \text{Beta}(s_1, s_2)$ truncated to $(0, \rho)$ (inverse-cdf method), $G \sim \text{Gamma}(s_1 + s_2, 1)$, and $\alpha_1 = BG/r_1$, $\alpha_2 = (1 - B)G/r_2$. The ordering can be checked from the unrestricted chain through $P(\alpha_{d1} < \alpha_{d2} \mid \text{data})$, the fraction of draws with $\alpha_1 < \alpha_2$.

4 Numerical Illustration

We now examine the performance of MLE and Bayes estimators of the Lomax parameters under JIAPCS, and evaluate the proposed estimates using absolute bias (AB, = |mean estimate − truth|) and mean squared error (MSE). Interval estimates are compared by average (median) width and coverage probability (CP). The Bayesian estimation algorithm and the simulation study are presented in the following subsection.

4.1 Simulation Study

In this section, we conduct a simulation study to evaluate the performance of classical and Bayesian estimators. Data are generated by Algorithm 2. We verified that the weights ν_{dj} add up to m_d in

Algorithm 2 Generation of JIAPCS competing risks data

1: For $d = 1, 2$ draw m_d uniform numbers U and set $X = \{(1-U)^{-1/(\alpha_{d1}+\alpha_{d2})} - 1\}/\lambda_d$ (inverse of S_d), sorted.
2: **for** $i = 1, 2, \ldots$ **do**
3: The next failure is the smaller of the two smallest surviving lifetimes. If it is $\geq T_2$, stop at T_2. Otherwise record it; if $i = k$ stop.
4: Use R_i if its time is $< T_1$ and 0 otherwise; withdraw at random R_i units from the population of the failed unit and $R_i + 1$ from the other.
5: **end for**
6: Withdraw all survivors at the stopping time; give each failure cause 1 with probability $\alpha_{d1}/(\alpha_{d1} + \alpha_{d2})$.

every replication.

Settings. True values: Ω_1: $(\alpha_{11}, \alpha_{12}, \lambda_1) = (0.5, 1.0, 1.0)$; Ω_2: $(\alpha_{21}, \alpha_{22}, \lambda_2) = (0.4, 0.8, 1.5)$ (so $\alpha_{d1} < \alpha_{d2}$). Designs $(m_1, m_2, k, \sum R_i)$: D1 $= (40, 35, 25, 6)$, D2 $= (60, 55, 40, 10)$, D3 $= (100, 90, 60, 20)$. Removal patterns: *early* ($R_1 = \sum R_i$), *mid* (split over the two middle failures) and *late* ($R_{k-1} = \sum R_i$). Thresholds (T_1, T_2) are the $(0.30, 0.60)$ (PA) or $(0.55, 0.90)$ (PB) quantiles of the lifetime law of Ω_1. This gives 18 main scenarios; further runs of design D2 (mid) use $\alpha_{d1} = \alpha_{d2}$ (boundary) and $\alpha_{d1} > \alpha_{d2}$ (restriction violated). Each scenario has 1000 replications; MCMC uses 800 burn-in and 2000 retained draws. MLE results are computed over the replications in which the MLE exists, Bayes results over all replications with $N_{d1}, N_{d2} \geq 1$; so the MLE is compared on an easier subset of samples, which favours the MLE.

Table 1: Designs (true values A, averaged over the 6 pattern/threshold combinations). Cases are I/II/III; the last column is the probability that a finite MLE exists in Ω_1/Ω_2.

Design	(m_1, m_2)	k	mean N_1/N_2	cases I/II/III	P(MLE exists)
D1	(40,35)	25	13.5 / 11.5	0.41 / 0.55 / 0.04	0.56 / 0.61
D2	(60,55)	40	20.7 / 19.1	0.30 / 0.60 / 0.10	0.65 / 0.71
D3	(100,90)	60	31.7 / 28.3	0.42 / 0.56 / 0.02	0.68 / 0.75

Table 2: Point estimation, design D2 (average over pattern/threshold scenarios). MLE: median error and MSE (conditional on existence); Bayes (SE loss): AB and MSE. U = unrestricted, R = order-restricted.

	MLE		WIP-U		WIP-R		IP-U		IP-R	
Parameter (true)	med. err.	MSE	AB	MSE	AB	MSE	AB	MSE	AB	MSE
α_{11} (0.5)	−0.17	41.2	0.208	0.092	0.156	0.055	0.022	0.019	0.003	0.013
α_{12} (1.0)	−0.33	202.0	0.325	0.186	0.376	0.206	0.044	0.029	0.069	0.026
λ_1 (1.0)	+0.66	14.4	0.032	0.060	0.032	0.060	0.051	0.020	0.051	0.021
α_{21} (0.4)	−0.10	48.5	0.320	0.157	0.261	0.101	0.020	0.016	0.005	0.010
α_{22} (0.8)	−0.20	194.9	0.548	0.388	0.605	0.433	0.044	0.027	0.068	0.025
λ_2 (1.5)	+0.64	29.5	0.358	0.198	0.357	0.198	0.046	0.028	0.046	0.027

Results for D1 and D3 are in the Appendix, and all scenario-level results are in the supplementary CSV files. The main findings are the following.

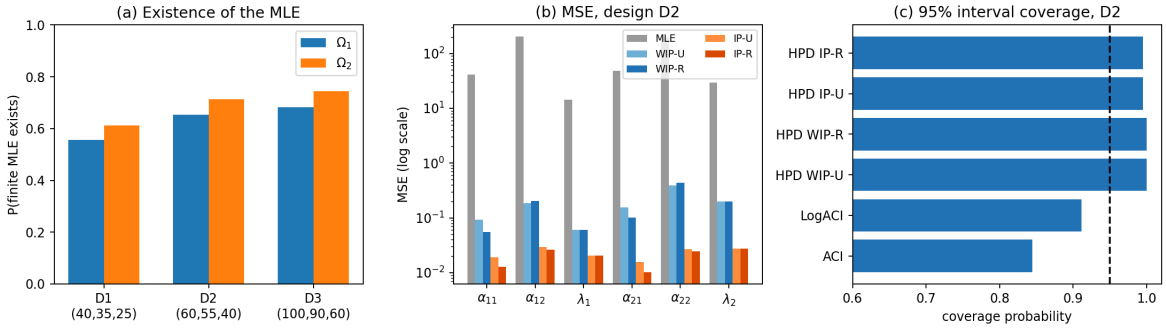


Figure 1: Simulation summary: (a) probability that a finite MLE exists; (b) MSE of MLE and Bayes (SE) estimates for D2; (c) coverage of 95% intervals for D2 (dashed: 0.95).

Table 3: MSE of Bayes estimates under different losses, D2, unrestricted. LINEX±: $p = \pm 0.5$; GE±: $q = \pm 0.5$.

		IP					WIP			
Par.	SE	LINEX+	LINEX−	GE+	GE−	SE	LINEX+	LINEX−	GE+	GE−
α_{11}	0.019	0.018	0.021	0.019	0.018	0.092	0.066	0.138	0.037	0.066
α_{12}	0.029	0.026	0.036	0.027	0.027	0.186	0.102	0.396	0.067	0.128
λ_1	0.020	0.017	0.028	0.019	0.018	0.060	0.045	0.144	0.084	0.053
α_{21}	0.016	0.015	0.017	0.015	0.015	0.157	0.118	0.221	0.062	0.116
α_{22}	0.027	0.024	0.031	0.024	0.025	0.388	0.245	0.693	0.150	0.287
λ_2	0.028	0.023	0.042	0.030	0.025	0.198	0.272	0.160	0.446	0.269

Table 4: 95% intervals, D2: median width and coverage probability.

	ACI		LogACI		HPD WIP-U		HPD WIP-R		HPD IP-U		HPD IP-R	
Par.	width	CP	width	CP	width	CP	width	CP	width	CP	width	CP
α_{11}	1.57	0.75	3.64	0.90	1.39	1.00	1.31	1.00	0.76	0.99	0.72	0.99
α_{12}	3.18	0.75	7.29	0.89	2.44	1.00	2.48	1.00	1.26	1.00	1.25	1.00
λ_1	10.13	1.00	47.05	0.92	2.35	1.00	2.34	1.00	1.42	1.00	1.42	1.00
α_{21}	1.32	0.78	2.80	0.92	1.43	1.00	1.32	1.00	0.62	0.98	0.58	0.98
α_{22}	2.62	0.78	5.22	0.91	2.49	1.00	2.53	1.00	1.02	1.00	1.01	1.00
λ_2	12.18	1.00	54.35	0.93	2.52	1.00	2.52	1.00	1.93	1.00	1.92	1.00

Table 5: Effect of the order restriction (D2; average over the four shape parameters): MSE of the SE Bayes estimate, median HPD width and HPD coverage, unrestricted (U) versus restricted (R).

	IP						WIP					
Truth	MSE U	MSE R	wid. U	wid. R	CP U	CP R	MSE U	MSE R	wid. U	wid. R	CP U	CP R
$\alpha_{d1} < \alpha_{d2}$ (true)	0.023	0.018	0.92	0.89	0.99	0.99	0.206	0.199	1.94	1.91	1.00	1.00
$\alpha_{d1} = \alpha_{d2}$ (boundary)	0.024	0.026	0.92	0.83	0.99	0.99	0.197	0.200	1.92	1.87	1.00	1.00
$\alpha_{d1} > \alpha_{d2}$ (violated)	0.022	0.091	0.91	0.80	0.99	0.91	0.200	0.265	1.92	1.84	1.00	1.00

Table 6: Removal pattern and thresholds (D2, true A): case frequencies I/II/III, mean N_1/N_2, mean probability that the MLE exists (average of Ω_1 and Ω_2), and average MSE over the six parameters of the SE Bayes estimates.

Pattern	Thr.	cases	N_1/N_2	P(exist)	MSE WIP-U / IP-U
early	PA	0.00/0.67/0.33	20.6/18.6	0.71	0.180 / 0.024
early	PB	0.36/0.64/0.00	21.3/18.7	0.71	0.187 / 0.024
mid	PA	0.00/0.75/0.25	20.7/18.8	0.71	0.178 / 0.024
mid	PB	0.49/0.51/0.00	21.0/19.0	0.71	0.184 / 0.024
late	PA	0.01/0.99/0.00	20.1/19.9	0.64	0.175 / 0.022
late	PB	0.96/0.04/0.00	20.4/19.6	0.62	0.178 / 0.022

1. **The Lomax MLE often does not exist.** A finite MLE exists in only 51–79% of the replications across scenarios (on average 56/61% in D1, 65/71% in D2 and 68/75% in D3 for Ω_1/Ω_2), and the probability grows only slowly with the sample size, because the condition $\Psi > 0$ is about the *shape* of the sample (heavy tail), not its size. When the MLE exists it is very unstable: for D2 the MSE is between 14 and 202 (Tab. 2), because $\hat{\alpha}$ and $\hat{\lambda}$ are nearly confounded along the ridge $\alpha\lambda =$ constant; the median errors are modest, but a few estimates are huge. For the same reason the MLE's MSE does not decrease with the design size (e.g. α_{12}: 111, 202, 397 in D1, D2, D3).
2. **Bayes estimates with proper priors are far better.** Even the weakly informative WIP prior (mean 1, not equal to the truth) reduces the MSE by two to three orders of magnitude (e.g. α_{12}: 202 for the MLE, 0.19 for WIP-U). The informative prior IP is better still (MSE 0.016–0.029) but is available only in simulation because it is centred at the truth. WIP is pulled towards its prior mean (AB up to 0.55), so its error is mostly bias.
3. **Loss functions.** Differences between SE, LINEX and GE are small for IP. For WIP they follow the direction of the prior bias: since the prior mean of the α's (1) exceeds the truth, the over-estimation-averse estimators (LINEX+, GE+) have smaller MSE for α, while for λ_2 (prior mean $1 < 1.5$) LINEX− is best (Tab. 3). Asymmetric losses help only when the likely direction of the error is known.
4. **Intervals.** The ACI under-covers for the shapes (CP 0.75–0.78) and has an absurd width for λ (median 10–12). The log-ACI is better (CP 0.89–0.93) but still below 0.95 and much wider than the HPD intervals. The HPD intervals have CP 0.98–1.00: they are conservative, because both priors are centred at moderate values and the likelihood is flat (Tab. 4, Fig. 1(c)). Bootstrap intervals were computed for the real data only, because 1000 bootstrap resamples per replication were too expensive for the whole design.
5. **Order restriction.** When $\alpha_{d1} < \alpha_{d2}$ is true, the restriction changes the MSE little on average (WIP: 0.206 → 0.199; IP: 0.023 → 0.018). For IP it lowers the MSE of all four shapes, but for WIP it helps the smaller shapes and hurts the larger ones (Tab. 2). When the restriction is violated, the MSE of the IP estimate quadruples (0.022 → 0.091), the WIP MSE rises from 0.200 to 0.265, and the HPD coverage of IP falls from 0.99 to 0.91 (Tab. 5). The restriction should therefore be used only when the ordering is justified, which can be checked with the LRT of Sec. 2 or with $P(\alpha_{d1} < \alpha_{d2} \mid \text{data})$.
6. **Censoring pattern.** Early and mid removals give practically identical results, while late removals lower the existence probability of the MLE slightly (0.62–0.64 versus 0.71), possibly because units withdrawn late (or after T_1, where the adaptive rule cancels removals) carry less information about the tail; we did not test this explanation further. The Bayes MSEs hardly depend on the pattern, and the thresholds mainly change the case mix (Tab. 6).

4.2 Real Data Analysis

In this section, we present an analysis of real data for illustrative purposes.

Aircraft air-conditioning failures (joint design). Proschan's [14] air-conditioning failure times (hours) for Boeing 720 planes 7914 (Ω_1, $m_1 = 24$) and 7913 (Ω_2, $m_2 = 27$) were tabulated with the failure cause (cause 1: compressor; cause 2: all others) by Chen and Gui [5]. (In that table the columns are described as lifetime, population, cause, but the values only fit the reading lifetime, cause, population with 24 and 27 units per plane; we use that reading.) In Ω_1 there are 15 compressor and 9 other failures; in Ω_2 there are 10 and 17.

Adequacy of the Lomax model. For the complete data the existence statistic is negative for both planes ($\Psi = -0.66$ and -3.50; squared coefficients of variation 0.91 and 0.66), so no finite MLE exists: the Lomax likelihood is maximised at the exponential limit and the data cannot distinguish a Lomax from an exponential model. The LRT of Section 2 for $\alpha_{d1} = \alpha_{d2}$ gives $\Lambda = 1.52$ ($p = 0.22$)

for Ω_1 (counts 15/9) and $\Lambda = 1.84$ ($p = 0.18$) for Ω_2 (counts 10/17): the two cause-specific shapes are not significantly different.

JIAPCS applied to the data. We applied the scheme with $k = 18$, $R_8 = R_9 = 3$ (all other $R_i = 0$; $\sum(R_i + 1) = 23 < 24$), $T_1 = 0.45$, $T_2 = 1.8$; the withdrawn units are chosen at random from the survivors. Over 200 random withdrawal patterns a finite MLE existed for Ω_1 in 47% and for Ω_2 in 20% of the patterns, and for both in only 5.5%. To illustrate the full procedure we report the first pattern (random seed 40) for which both MLEs exist; this is an *atypical* sample chosen for illustration (Case II, 18 failures: Ω_1 with $N = 10$, causes 5/5; Ω_2 with $N = 8$, causes 6/2). Results are in Tab. 7 and Fig. 2.

Table 7: Aircraft data (one JIAPCS sample). MLE, restricted MLE (R), bootstrap-p 95% interval, and Bayes estimates under the WIP prior: SE, LINEX ($p = 0.5$), GE ($q = 0.5$) with 95% HPD interval, and the restricted SE estimate with HPD.

Par.	MLE	MLE-R	Boot-p	SE	LINEX	GE	HPD	SE-R	HPD-R
α_{11}	1.631	1.631	[0.10, 11.59]	1.118	1.018	0.857	[0.19, 2.50]	0.875	[0.15, 1.84]
α_{12}	1.631	1.631	[0.08, 11.70]	1.107	1.008	0.847	[0.17, 2.50]	1.378	[0.33, 2.85]
λ_1	0.539	0.539	[0.06, 19.44]	1.109	0.999	0.834	[0.17, 2.50]	1.060	[0.19, 2.42]
α_{21}	0.946	0.630	[0.06, 8.11]	1.205	1.076	0.902	[0.18, 2.82]	0.763	[0.10, 1.70]
α_{22}	0.315	0.630	[0.03, 3.46]	0.520	0.480	0.328	[0.03, 1.33]	1.028	[0.17, 2.32]
λ_2	0.762	0.762	[0.08, 36.57]	0.834	0.749	0.567	[0.06, 2.03]	0.799	[0.08, 2.03]

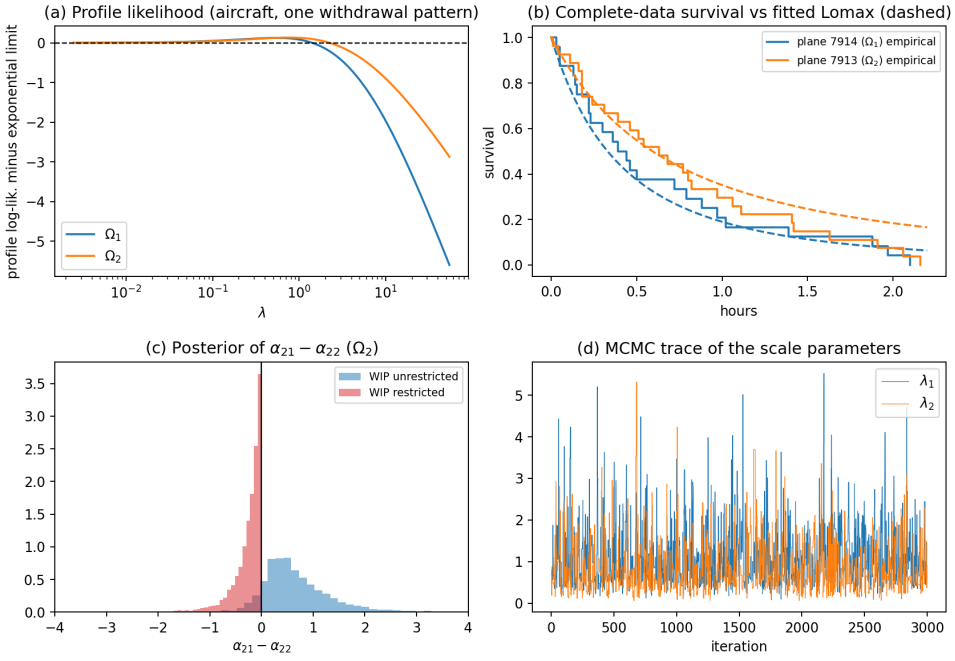


Figure 2: Aircraft data: (a) profile log-likelihood relative to the exponential limit; (b) empirical survival functions of the complete samples and fitted Lomax curves; (c) posterior of $\alpha_{21} - \alpha_{22}$; (d) MCMC traces of λ_1, λ_2.

Findings. (i) The profile log-likelihood exceeds its exponential-limit value by only 0.13 for both populations (Fig. 2(a)). As an LRT of "exponential" against "Lomax" this is $2 \times 0.13 = 0.26$; since the exponential model sits on the boundary ($\alpha \to \infty$), the null distribution is the mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ and the approximate p-value is $\frac{1}{2}P(\chi^2_1 > 0.26) \approx 0.31$. There is no evidence for a Lomax shape. (ii) The bootstrap-p intervals are extremely wide (e.g. λ_1: $[0.06, 19.4]$) and only 52% and 45% of the bootstrap samples had a finite MLE. (iii) The Bayes estimates are close to the prior means (WIP has mean 1), so the data hardly update the prior; the HPD intervals are

wide. (iv) In this particular censored sample Ω_2 has 6 failures of cause 1 and only 2 of cause 2, so the unrestricted posterior gives $P(\alpha_{21} < \alpha_{22} \mid \text{data}) = 0.09$, whereas in the complete data cause 2 is more frequent (17 versus 10). The restricted estimate simply imposes the ordering, which shows why the restriction should be checked and why very small samples cannot support it. (v) The fitted Lomax curves follow the empirical survival curves reasonably well (Fig. 2(b)), as they must when the model is close to exponential.

Electrical appliances (single-population analysis). Because the aircraft data are near-exponential, we also used a data set with a clearly decreasing early failure rate: the 36 small electrical appliances of Lawless [15], in the improved-adaptive competing risks form used in the literature (see [8]). The first 16 failures ($n = 36$) occurred at times $11, 35, 49, 170, \ldots, 2400$ (we divided by 1000) and the remaining 20 units were withdrawn at the last failure, i.e. $\nu = (1, \ldots, 1, 21)$. Cause 2 is failure mode 9 (10 failures) and cause 1 collects all other modes (6 failures); the labels are chosen so that $\alpha_1 \le \alpha_2$ is the plausible ordering. Since the likelihood factorises by population, this analysis uses exactly the same estimators (a JIAPCS in which only one population is observed); we do *not* claim a joint analysis here. Results are in Tab. 8 and Fig. 3.

Table 8: Appliance data: estimates as in Tab. 7.

Par.	MLE	MLE-R	Boot-p	SE	LINEX	GE	HPD	SE-R	HPD-R
α_1	0.602	0.602	[0.08, 6.54]	0.533	0.490	0.352	[0.05, 1.39]	0.463	[0.05, 1.20]
α_2	1.004	1.004	[0.14, 10.18]	0.831	0.747	0.577	[0.10, 2.09]	0.844	[0.12, 2.13]
λ	0.181	0.181	[0.01, 2.90]	0.471	0.422	0.260	[0.02, 1.39]	0.489	[0.04, 1.51]

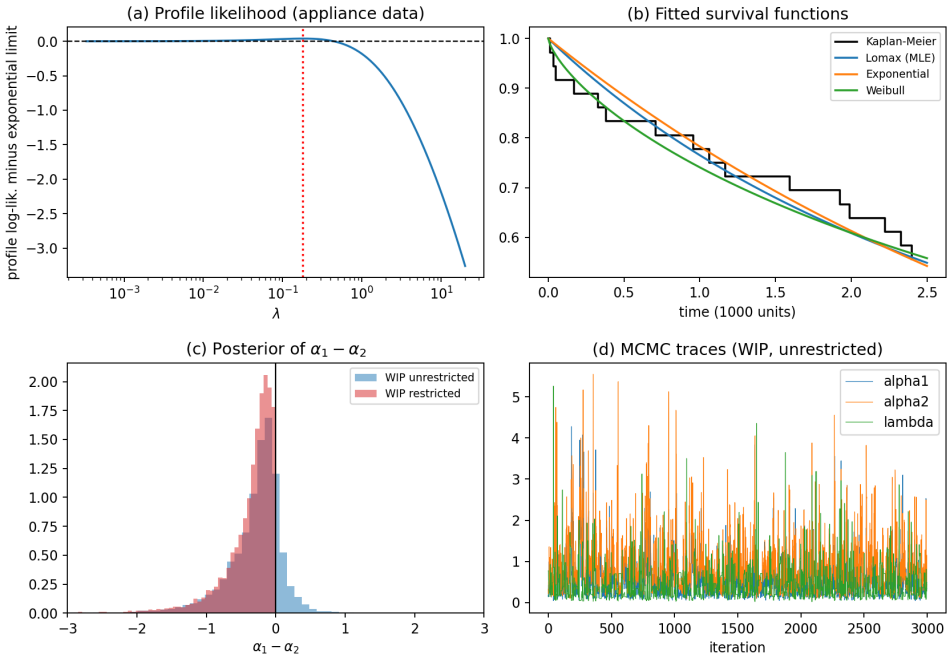


Figure 3: Appliance data: (a) profile log-likelihood relative to the exponential limit (dotted: MLE); (b) Kaplan–Meier curve and fitted survival functions; (c) posterior of $\alpha_1 - \alpha_2$; (d) MCMC traces.

Findings. Here $\Psi = 0.52 > 0$ and the MLE exists: $\hat{\alpha} = (0.602, 1.004)$, $\hat{\lambda} = 0.181$. (i) The asymptotic intervals are useless (ACI for α_1: $[-3.7, 4.9]$; log-ACI: $[0.0005, 763]$), confirming the simulation; the bootstrap-p intervals are wide but sensible, and only 47% of the bootstrap samples had a finite MLE. (ii) Ignoring the cause labels, the censored log-likelihoods of the lifetime distri-

bution are -38.47 (Lomax, 2 parameters), -38.51 (exponential, 1) and -37.48 (Weibull, 2), so the AICs are 80.9, 79.0 and 79.0. The Lomax therefore gives no real improvement over the exponential and is not preferred by AIC, although all three curves follow the Kaplan–Meier curve closely (Fig. 3(b)). The LRT against the exponential gives $2 \times 0.04 \approx 0.08$ and an approximate boundary p-value of about 0.39. (iii) The WIP posterior (SE) is $(0.533, 0.831, 0.471)$ and a Gamma$(2, 2)$ prior gives nearly the same $(0.497, 0.749, 0.425)$, whereas the NIP prior gives posterior means of α of 40.7 and 70.7 (Remark 3.1 in action). (iv) $P(\alpha_1 < \alpha_2 \mid \text{data}) = 0.83$, so the restriction is plausible but not strongly supported, and the restricted and unrestricted estimates are close.

5 Conclusion

In this paper, we develop classical and Bayesian estimators of the Lomax competing risks model under the joint improved adaptive progressive censoring scheme. The likelihood factorises by population, the shapes have closed-form conditional MLEs, the order-restricted MLE is obtained by pooling, and a collapsed sampler gives exact Bayes estimates for the unrestricted and the order-restricted prior. Our main conclusions are:

- The Lomax MLE exists only when the sample is heavy-tailed enough ($\Psi > 0$); otherwise the data support the exponential limit. A finite MLE existed in only 51–79% of simulated samples and was very unstable, so its existence should always be checked before it is reported.
- Bayes estimates with proper (even weakly informative) priors are much more accurate and always exist; vague priors with tiny hyperparameters must be avoided for this model.
- Asymptotic intervals under-cover; HPD intervals are shorter and conservative.
- The order restriction is useful only if it is true; its plausibility can be checked with the likelihood ratio test or the posterior probability of the ordering.
- Both real data sets were close to exponential, so they illustrate the procedures but cannot confirm a Lomax shape.

Limitations. We assumed a common λ_d for the two causes within a population and independent latent lifetimes; the simulation uses one configuration of true values; the informative prior is centred at the truth and is therefore optimistic; the MLE is evaluated only on the samples where it exists; the bootstrap intervals were not part of the simulation study; uniqueness of $\hat{\lambda}$ was checked numerically, not proved; and the aircraft illustration uses a rare withdrawal pattern. Natural extensions are different scale parameters for the two causes, bootstrap-t intervals, optimal choice of the censoring scheme, and dependent causes.

A Results for designs D1 and D3

Table 9: Point estimation, design D1 (same layout as Tab. 2).

	MLE		WIP-U		WIP-R		IP-U		IP-R	
Parameter (true)	med. err.	MSE	AB	MSE	AB	MSE	AB	MSE	AB	MSE
α_{11} (0.5)	-0.23	118.2	0.229	0.112	0.144	0.050	0.024	0.022	0.013	0.013
α_{12} (1.0)	-0.45	111.2	0.311	0.182	0.397	0.215	0.041	0.034	0.078	0.030
λ_1 (1.0)	$+1.21$	54.8	0.035	0.053	0.034	0.054	0.051	0.025	0.051	0.025
α_{21} (0.4)	-0.17	44.1	0.349	0.188	0.248	0.093	0.019	0.018	0.018	0.011
α_{22} (0.8)	-0.33	291.5	0.536	0.384	0.637	0.468	0.037	0.032	0.074	0.029
λ_2 (1.5)	$+1.44$	186.8	0.375	0.200	0.375	0.200	0.042	0.030	0.042	0.030

Table 10: Point estimation, design D3.

	MLE		WIP-U		WIP-R		IP-U		IP-R	
Parameter (true)	med. err.	MSE	AB	MSE	AB	MSE	AB	MSE	AB	MSE
α_{11} (0.5)	−0.15	85.8	0.200	0.078	0.172	0.057	0.023	0.015	0.009	0.011
α_{12} (1.0)	−0.28	397.3	0.339	0.185	0.367	0.196	0.047	0.024	0.062	0.022
λ_1 (1.0)	+0.54	7.48	0.028	0.063	0.028	0.063	0.054	0.018	0.054	0.018
α_{21} (0.4)	−0.10	24.8	0.305	0.135	0.271	0.103	0.019	0.012	0.004	0.009
α_{22} (0.8)	−0.19	85.3	0.550	0.377	0.583	0.404	0.043	0.021	0.057	0.019
λ_2 (1.5)	+0.65	13.1	0.349	0.193	0.349	0.193	0.046	0.025	0.047	0.025

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